

Akanksha

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EXPERIENCE

CSIR-CSIO CHANDIGARH

Research Trainee

Chandigarh

Jan 2024 – Jul 2024

- Optimized YOLOv7 architecture by studying and integrating advanced models (transformers, LSTMs, attention mechanisms, Swin Transformer), achieving a lightweight model with a 54% reduction in training time (2.2× faster).
- Created detailed Jupyter notebooks illustrating stepwise model workings, enabling informed decision-making on architectural tweaks. Achieved a 5% increase in accuracy while significantly reducing model parameters, enabling efficient training on a single GPU within one day.
- Co-authored a research paper** documenting these architectural advancements, published in the *International Conference on Pattern Recognition (ICPR), 2024*. [[Link](#)]

EDUCATION

PEC UNIVERSITY OF TECHNOLOGY

Bachelor of Technology (B.Tech) in Aerospace Engineering | Cumulative GPA: 7.9/10

Chandigarh

Nov 2021 - May 2025

DR DEVI CHAND D.A.V. PUBLIC SCHOOL, MOHAL

Senior Secondary Education (12th) | Percentage - 95.6%

High School Education (10th) | Percentage - 96.2%

Kullu, Himachal Pradesh

Apr 2020 - Mar 2021

Apr 2018 - Mar 2019

SKILLS

Programming Languages: Python, SQL

Libraries & Frameworks: PyTorch, OpenCV, Huggingface, Pandas, NumPy, Matplotlib, Seaborn, LangChain

ML & Deployment Tools: ZenML, MLflow, Streamlit, Flask, Git

Languages: English (Professional), Hindi (Native), German (Beginner), Japanese (Beginner)

PROJECTS

Flight Price Prediction

- Engineered an end-to-end ML pipeline to predict flight prices using a Kaggle dataset, implementing feature engineering and training models with Scikit-Learn.
- Automated training and evaluation with ZenML & MLflow, and prepared the model for deployment.

TED Talk Recommender

- Built a content-based recommender using YouTube Data API, SQLite, and Hugging Face embeddings to suggest relevant TED Talks.
- Developed dual UIs with Streamlit (deployed on Streamlit Cloud) and Flask for end-to-end ML deployment practice.

Urban Noise Clustering

- Designed an end-to-end ML pipeline to analyze and cluster NYC 311 noise complaints.
- Implemented feature engineering, baseline & ensemble clustering models, and a Flask dashboard for visualization.

EXTRA-CURRICULAR

Executive Body Member, Music Club, PEC

Nov 2022 - May 2025

- Organized and coordinated 40+ music events, including the Solo Singing Competition at **PECFEST 2024** and **2025**.
- Won second prize in Battle Of Bands organized by BITS Pilani, competing against 10 teams.
- Earned the COA in 2022-23 and COE for contributions during the 2023-24 and 2024-25 sessions, recognizing the top 6% of members for leadership and event impact.

Marketing And Creative Team Member, VONG - Voice Of New Generation

Nov 2023 - Feb 2024

- Directed a marketing team of five members in Dec 2023 and collaborated with two cross-functional teams to deliver campaigns, expecting an increase in engagement by 10% within three months.